MODULE-WISE QUESTION BANK

Academic Year :	2026-27	**Batch :**	2024
Course Code :	BCS502	**Course Title :**	Computer Network and Security
Credits :	4	**Course Coordinator :**	Prof. Nagamani D R
Total Contact Hours :	**40L+20P**	**Total Hours of Pedagogy:**	40+20
Faculty In-charge :	Dr. GHS, Prof. DRN, & Prof. KNP	**Semester & Section**	5th (A, B, C & D)

MODULE - 1: Introduction, Network Models, Error Detection and Correction, Media Access Control

Q. No.	Question	CO	RBL
1	Discuss the fundamental characteristics of data communication; With neat diagram explain components of Network communication system.	1	L2
2	Discuss the different modes of communication between two devices with suitable diagrams and examples.	1	L2
3	Explain different types of Topologies in communication networks with Pros and cons.	1	L2
4	With neat layer diagram of TCP/IP protocol suite with identical objects and logical connections at sender and receiver side, and briefly discuss the services of each layer.	1	L2
5	With neat diagram, explain Encapsulation at source and Decapsulation at destination and both at Router.	1	L2
6	Illustrate the concept of Multiplexing and de multiplexing at the upper layers.	1	L2
7	Classify the types of errors encountered in computer networks and explain block coding.	1	L2
9	Analyze two code words to determine their Hamming distance and calculate the minimum Hamming distance of a given code.	4	L3
10	Construct the encoder and decoder operations for a simple parity-check code for a given data word.	4	L3
11	Define redundancy. Explain CRC encoder and decoder with a block diagram.	4	L3
	Explain Fletcher checksum, and Adler checksum with flowchart. Calculate the checksum for the following bytes $(2B)_{16}$,$(3F)_{16}$,$(6A)_{16}$ and $(AF)_{16}$. using Fletcher algorithm, Also show that the result is a weighted checksum		
12	Explain working of traditional checksum with flowchart, Suppose that the sender wants to send 4 frames each of 8 bits, where the frames are 11001100, 10101010, 11110000 and 11000011 Find the checksum at sender side and also check at receiver weather the receiver received correct frames or not.	3	L3

13	Explain the operation of Pure ALOHA and Slotted ALOHA. Derive their vulnerable times and throughput expressions A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces a. 1000 frames per second? b. 500 frames per second? c. 250 frames per second?	2	L3
14	Explain and compare CSMA/CD and CSMA/CA, highlighting their collision-handling mechanisms and applications in wired and wireless networks.	2 2	L2
15	Explain the three controlled-access methods—reservation, polling, and token passing—and compare their suitability for different network environments.	2	L2
16	Solve the following: sender transmits 1101011011, using a generator 10011. During transmission, some bits are changed. a. Generate the CRC codeword. b. Determine the CRC at the receiver. c. Explain how the receiver identifies the error.	4	L3
17	Find the code word using CRC for given data word 1001 and divisor polynomial is $x^3 + x + 1$.	4	L3
19	A network contains hundreds of devices that frequently compete for a shared communication channel. How would you decide between random-access and controlled-access protocols? Justify your choice.	2	L3
20	**Case Study 1 – Hospital Network** A hospital wants to connect computers in the reception, laboratory, pharmacy, ICU and administration departments. The network must provide high reliability and security. a. Recommend a suitable topology. b. Identify the important network criteria. c. Explain the addressing mechanisms involved. d. Recommend an appropriate error-detection technique.	4	L4
22	**Case Study 2 – Online Classroom** An online classroom experiences delay, irregular audio and occasional packet loss. a. Identify the data communication characteristics affected. b. Explain the role of timeliness and jitter.	4	L4

	c. Suggest measures to improve communication quality.		

Faculty In-charge

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**